

ARCHITECTURE DOCUMENT

Project Overview

The Telecom KPI Anomaly Detection Dashboard is an AI-powered application developed to automatically detect abnormal behavior in telecom network Key Performance Indicators (KPIs). The project processes raw telecom KPI data, performs data preprocessing and feature engineering, scales the data using StandardScaler, applies the Isolation Forest machine learning algorithm to identify anomalous network conditions.

The detected anomalies are visualized through an interactive Streamlit Dashboard, enabling network engineers to monitor network performance, analyze KPIs, and download anomaly reports for further investigation.

Problem Statement

Telecommunication networks generate a massive amount of KPI data every day from thousands of base stations. Monitoring this data manually is time-consuming and prone to errors. Delayed detection of abnormal network behavior can result in service degradation, reduced customer satisfaction, and increased operational costs.

An automated anomaly detection system is required to identify unusual network behavior quickly and accurately, allowing engineers to take proactive corrective actions.

Business Importance

This project provides several benefits to telecom operators:

- Automates anomaly detection
- Reduces network downtime
- Improves overall network reliability
- Enables proactive network maintenance
- Reduces manual monitoring effort
- Improves customer experience
- Helps engineers identify abnormal KPI patterns quickly

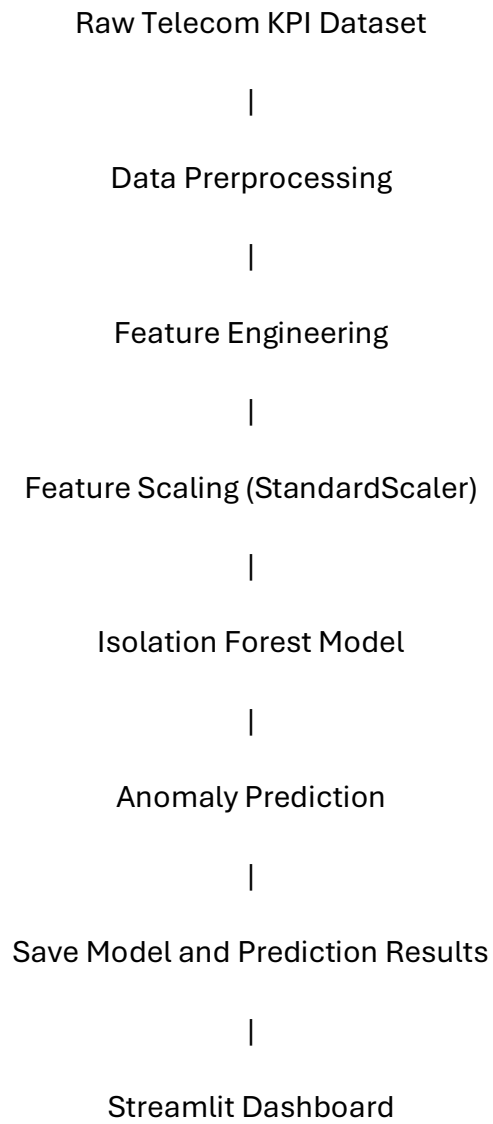
Project Objectives

The objectives of this project are:

- Preprocess telecom KPI data
- Handle missing values and duplicate records
- Engineer meaningful telecom network features
- Scale numerical features using StandardScaler
- Detect anomalies using Isolation Forest algorithm
- Display results through an interactive Streamlit dashboard
- Allow users to download anomaly detection results

System Architecture

The overall system architecture of the project is shown below



|

Network Engineer

Technology Stack

- Python 3.12
- VS Code
- Pandas
- Scikit-learn
- StandardScaler
- Isolation Forest
- Plotly Express
- Streamlit
- Joblib
- Git
- GitHub

Project Structure

telecom-ai-capstone/

├─ app/

| └─ streamlit_app.py

├─ data/

| └─ raw/

| └─ processed/

├─ docs/

├─ models/

├─ reports/

└─ screenshots/

```
├── src/
|   ├── preprocessing.py
|   ├── feature_engineering.py
|   ├── model.py
|   ├── evaluation.py
|   └── hyperparameter_tuning.py
├── README.md
├── requirements.txt
├── LICENSE
└── .gitignore
```

Folder Description

Folder	Description
app	Contains the Streamlit dashboard
data	Stores raw and processed datasets
docs	Project documentation
models	Saved Isolation Forest model and scaler
notebooks	Experimental notebooks
reports	Evaluation and tuning reports
screenshots	Dashboard screenshots
src	Source code modules

Data Flow

The data flows through the system as follows:

Raw Dataset-->Data Preprocessing-->Feature Engineering-->Feature Scaling-->Isolation Forest Model-->Anomaly Prediction-->Prediction Results-->Streamlit Dashboard -
->Download CSV Report

Model Pipeline

The machine learning pipeline consists of the following steps:

- Load the telecom KPI dataset
- Validate the dataset
- Clean missing values and duplicates
- Engineer new telecom features
- Scale numerical features using StandardScaler
- Train the Isolation Forest model
- Predict anomalous network records
- Save the trained model and predictions
- Display the results through the Streamlit dashboard

Feature Engineering

Three new features were created to improve anomaly detection performance.

Signal Quality Score

Formula: $0.6 * \text{SINR} + 0.4 * (\text{RSRP} + 120)$

Purpose:

Combines signal strength and signal quality into a single feature

Network Load

Formula: $\text{Connected Users} / 200$

Purpose:

Measures the percentage of network capacity currently being utilized

Network Health Index

Formula: $\text{Throughput} - \text{Latency} - (10 * \text{Packet Loss})$

Purpose:

Provides a combined measure of overall network performance

Machine Learning Model

The project uses the Isolation Forest algorithm for anomaly detection

Why Isolation Forest?

- Unsupervised machine learning algorithm
- Does not require labelled training data
- Efficient for large datasets
- Suitable for high-dimensional telecom KPI data
- Detects abnormal behavior by isolating outlier observations

Model Parameters

Parameter	Value
Algorithm	Isolation Forest
Number of Trees	100
Contamination	0.05
Random State	42

Dashboard

The Streamlit dashboard provides an interactive interface for monitoring telecom KPIs.

Dashboard Features

- Cell ID filtering
- KPI summary cards
- Throughput trend visualization
- SINR distribution
- RSRP vs Throughput scatter plot
- Anomaly distribution pie chart
- Download prediction results

Future Enhancements

Future improvements that can be added to the project include:

- Real time telecom KPI monitoring
- Automatic email/SMS alert notifications
- Root Cause Analysis (RCA)
- Cloud deployment on AWS, Azure, or GCP
- Live database integration
- User authentication and role-based access
- Comparison with other anomaly detection algorithms such as One-Class SVM and Autoencoders

Conclusion

The Telecom KPI Anomaly Detection Dashboard successfully demonstrates the use of machine learning for automated telecom network monitoring. By combining data preprocessing, feature engineering, StandardScaler, and the Isolation Forest algorithm, the system effectively detects anomalous KPI records and presents the results through an intuitive Streamlit dashboard. The solution improves monitoring efficiency, supports proactive maintenance, and provides a strong foundation for future enhancements such as real-time analytics and cloud deployment.